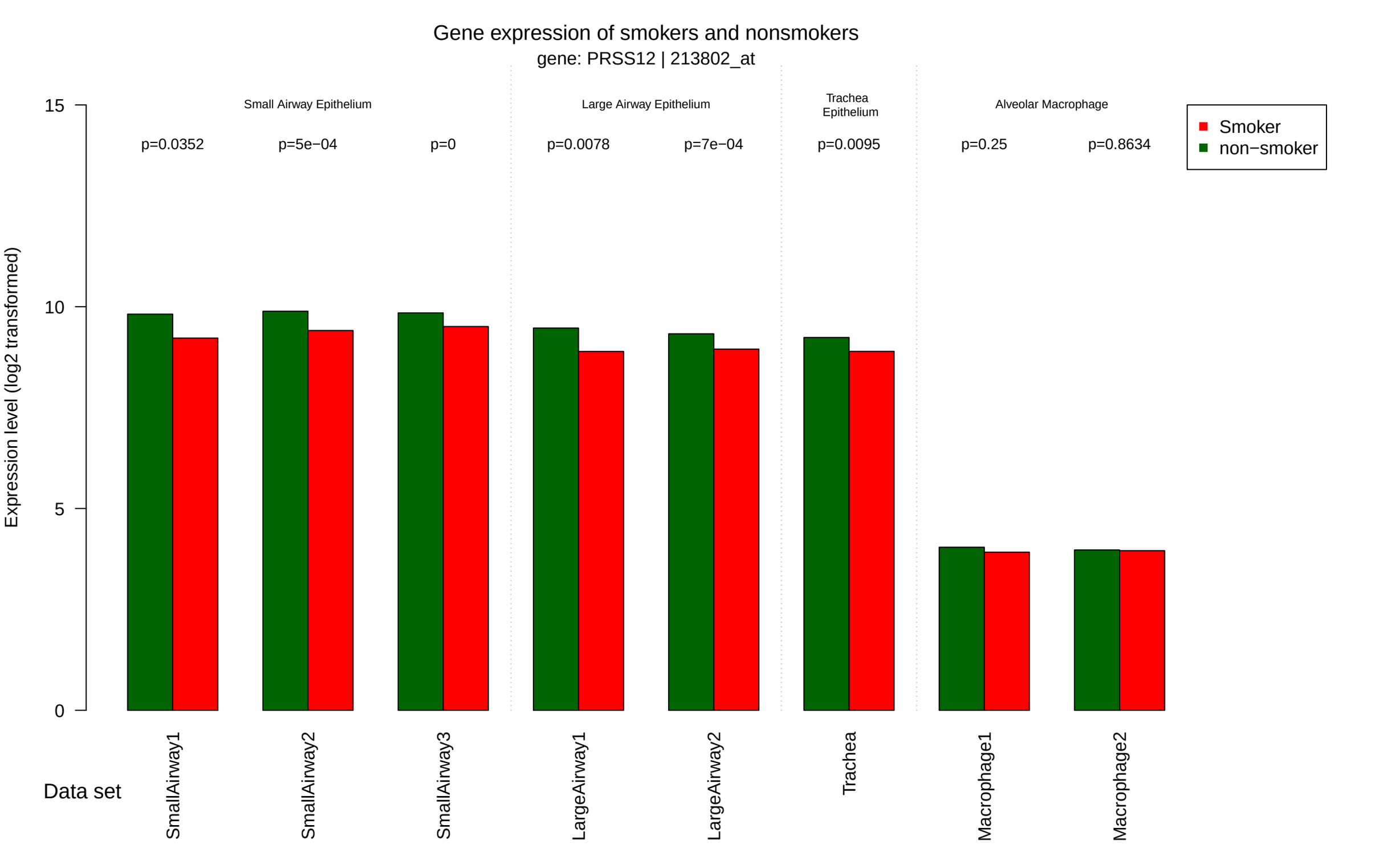
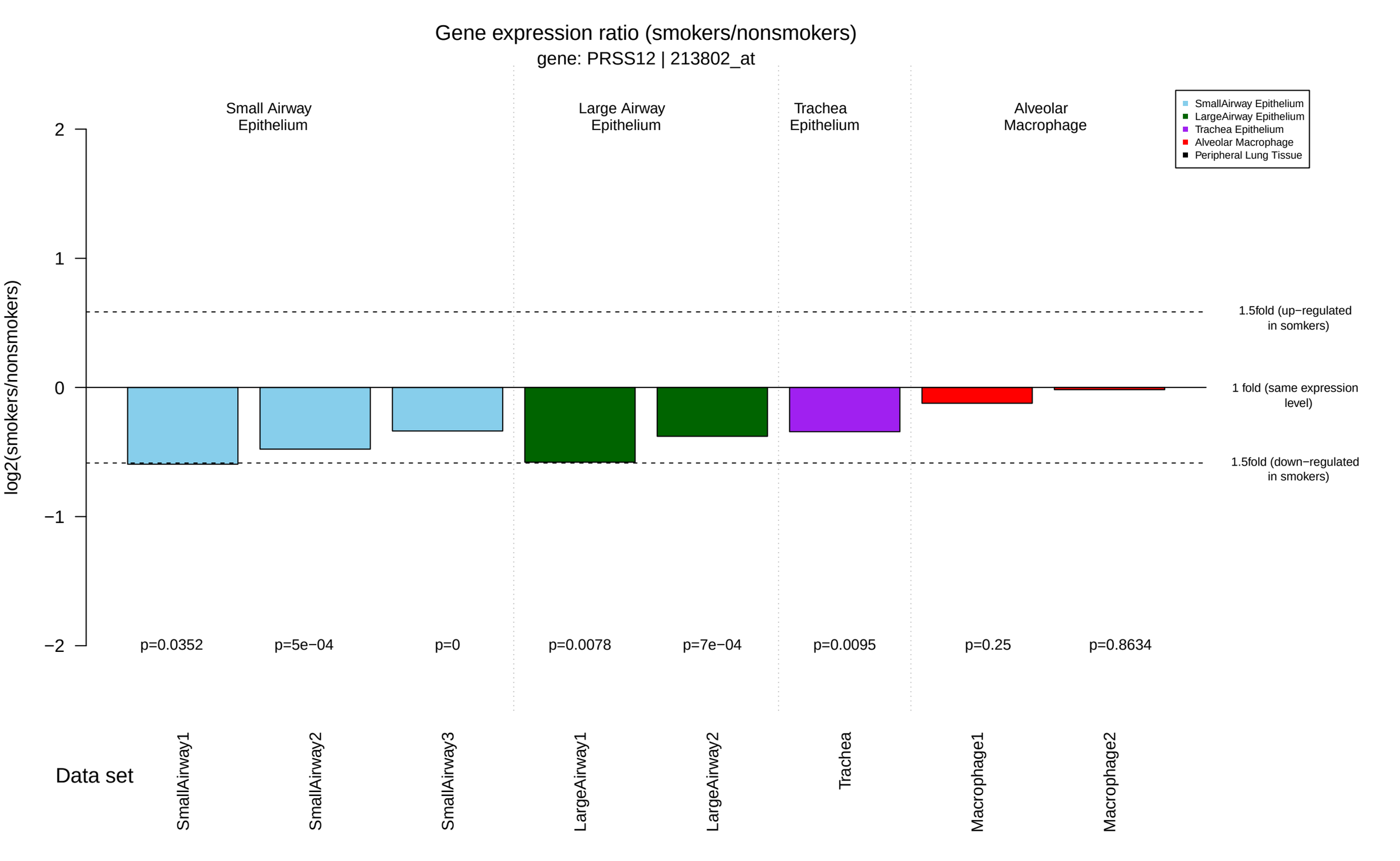
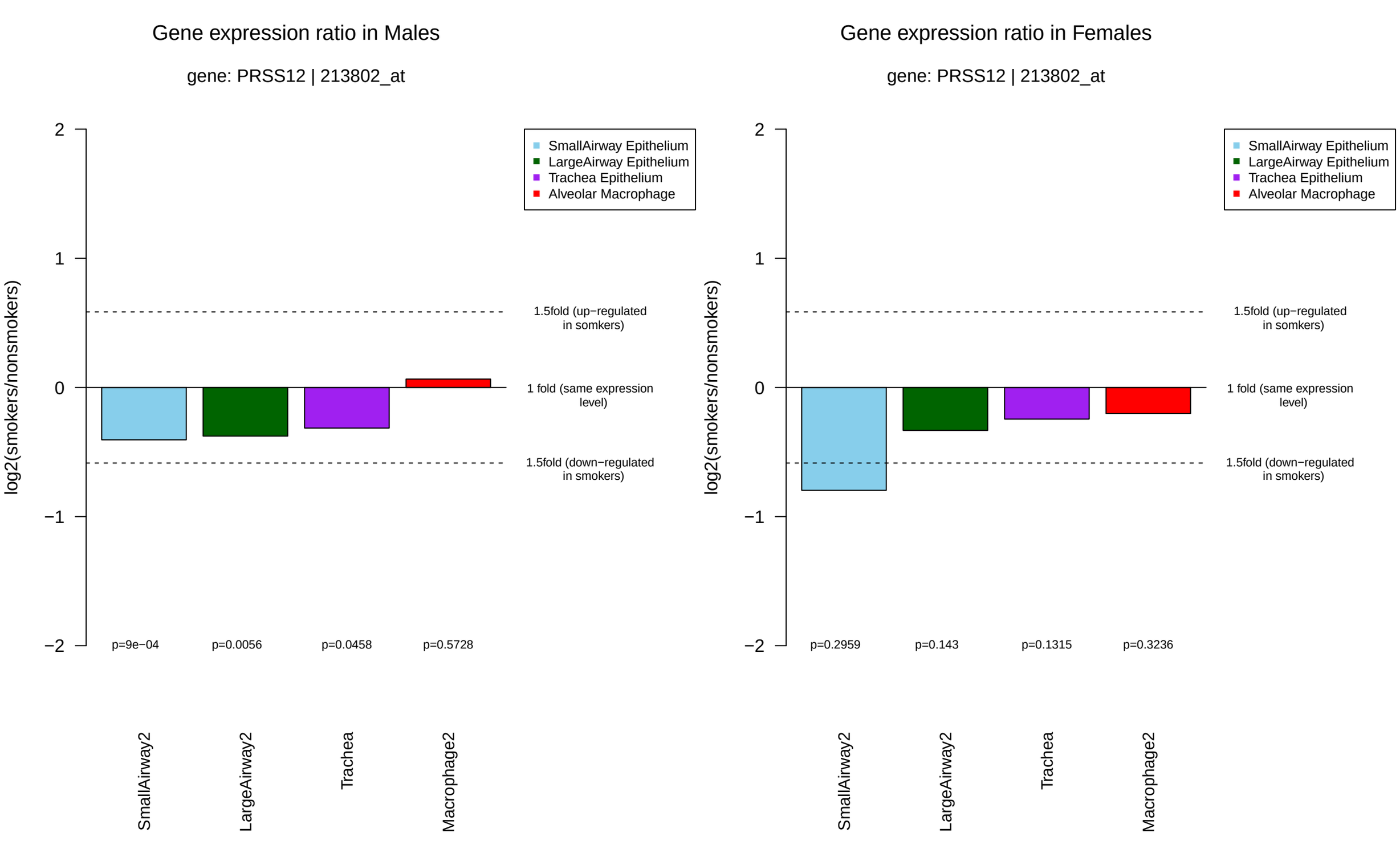
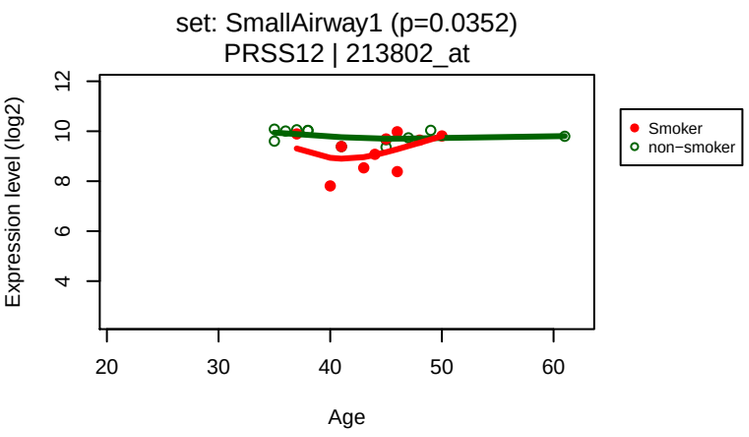
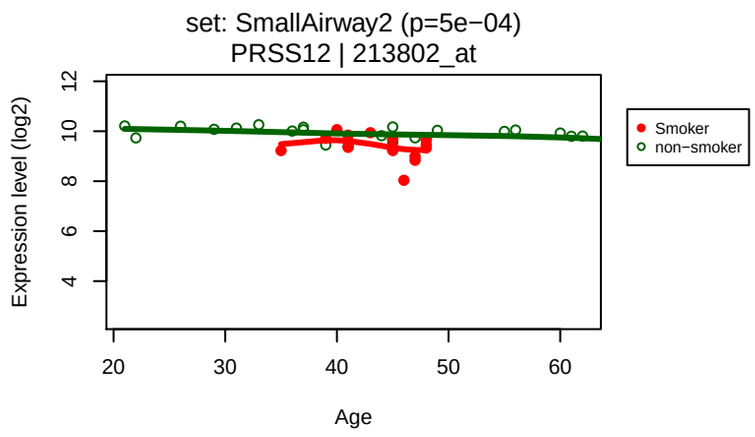
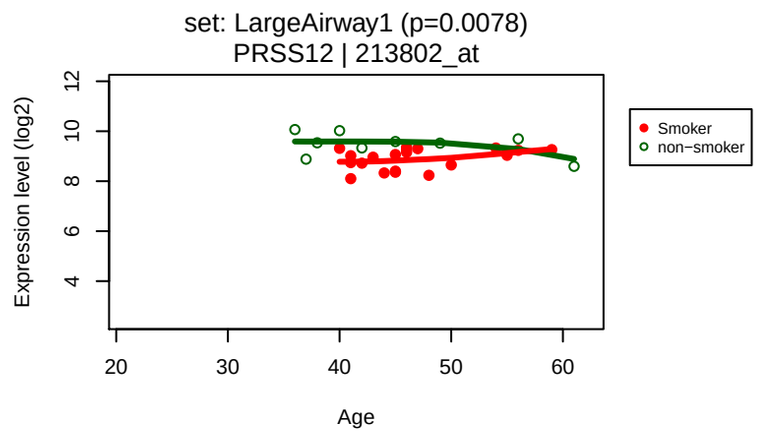

Gene expression of smokers and nonsmokers
gene: PRSS12 | 213802_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0352
p=5e−04
p=0
p=0.0078
p=7e−04
p=0.0095
p=0.25
p=0.8634
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

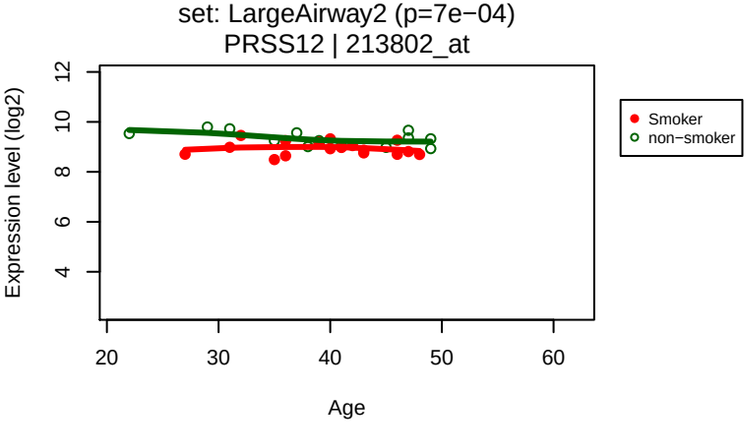

Gene expression ratio (smokers/nonsmokers)
gene: PRSS12 | 213802_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0352
p=5e−04
p=0
p=0.0078
p=7e−04
p=0.0095
p=0.25
p=0.8634
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

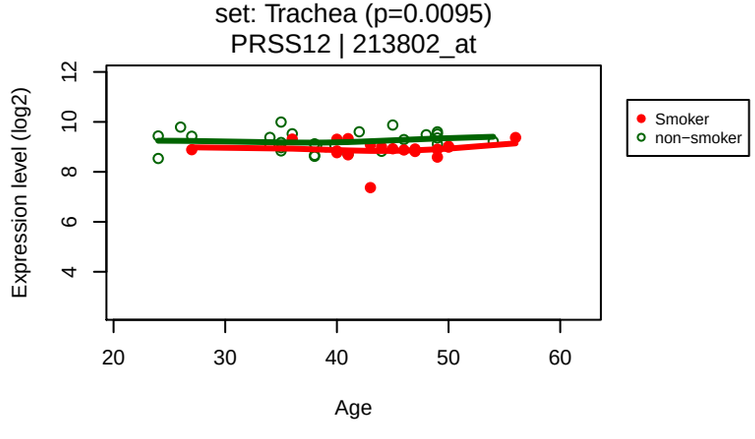

Gene expression ratio in Males
gene: PRSS12 | 213802_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=9e−04
p=0.0056
p=0.0458
p=0.5728
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS12 | 213802_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2959
p=0.143
p=0.1315
p=0.3236
SmallAirway2
LargeAirway2
Trachea
Macrophage2

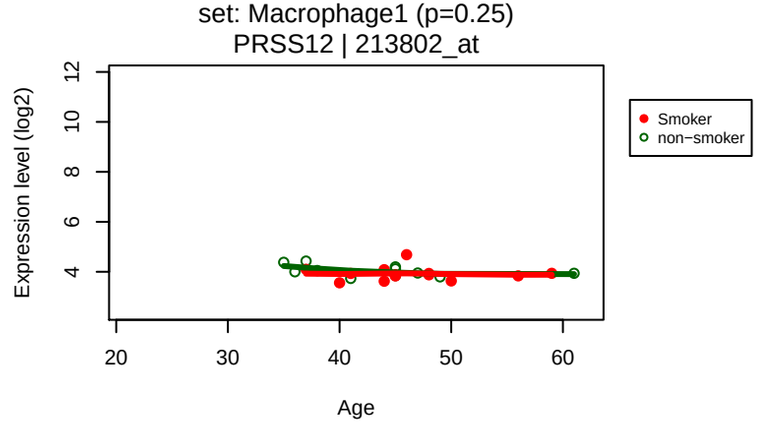

set: SmallAirway1 (p=0.0352)
PRSS12 | 213802_at
Expression level (log2)
Age
Smoker
non−smoker

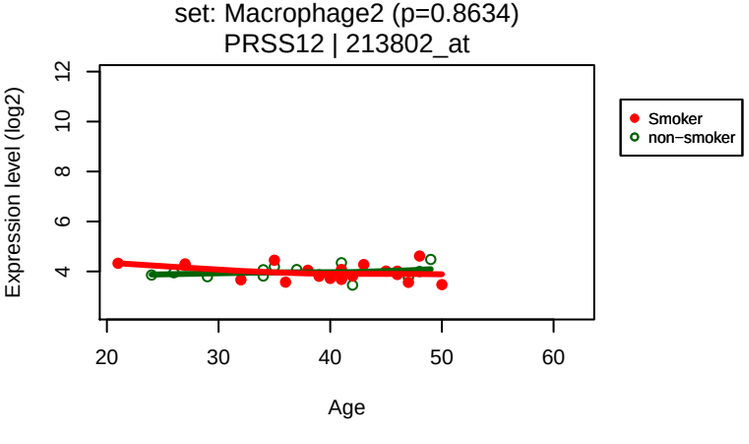

set: SmallAirway2 (p=5e−04)
PRSS12 | 213802_at
Expression level (log2)
Age
Smoker
non−smoker

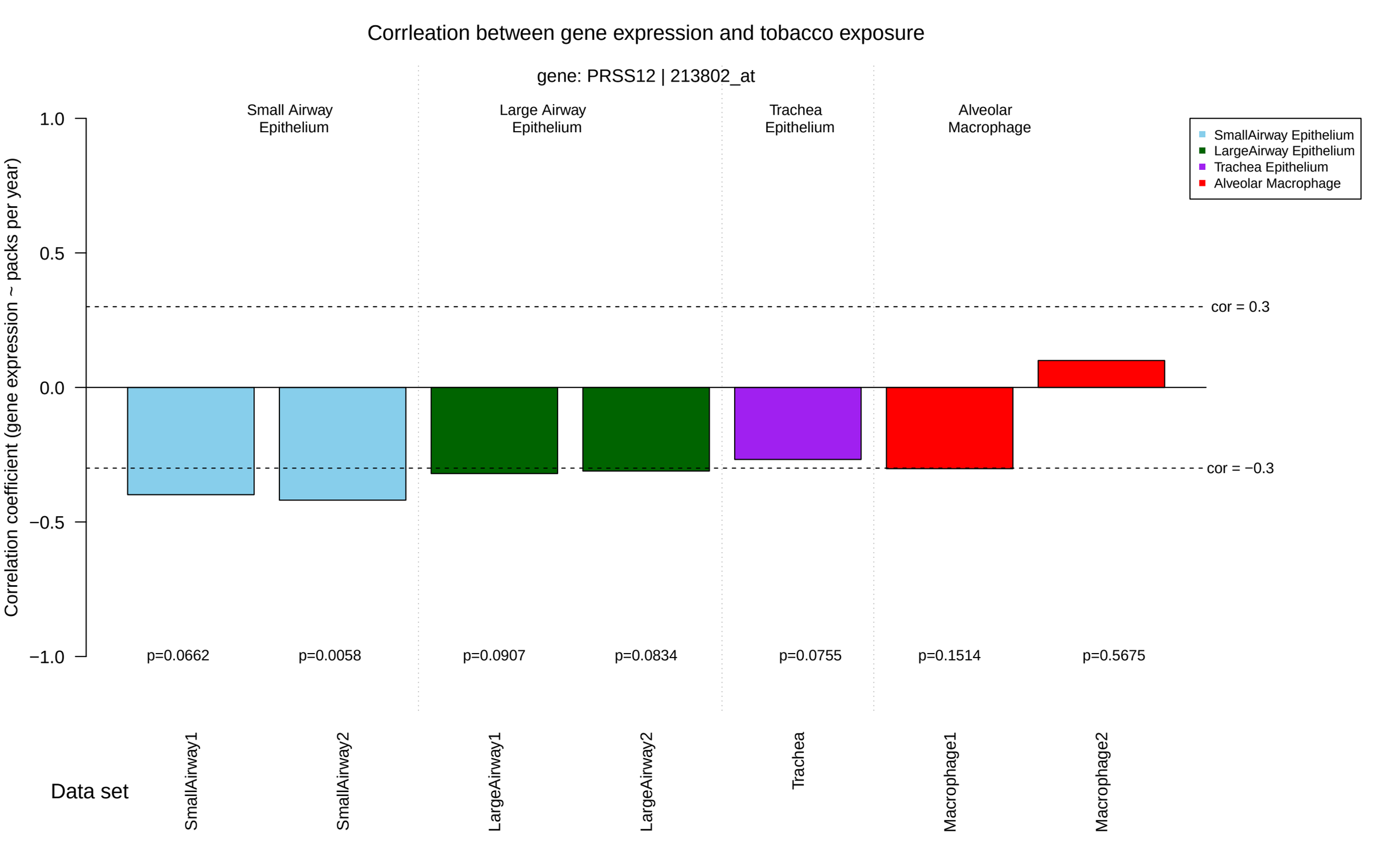

set: LargeAirway1 (p=0.0078)
PRSS12 | 213802_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=7e−04)
PRSS12 | 213802_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0095)
PRSS12 | 213802_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.25)
PRSS12 | 213802_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8634)
PRSS12 | 213802_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS12 | 213802_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0662
p=0.0058
p=0.0907
p=0.0834
p=0.0755
p=0.1514
p=0.5675
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2